

SERGEY ZINCHENKO [+44 7440 195498](tel:+447440195498) | zinch.s.e@gmail.com | linkedin.com/in/zinchse

SOFTWARE ENGINEER | Efficient Systems & ML | International Cryptography Gold Medalist & National Math 1st-Degree Awardee

SUMMARY: Engineer with **4+** years of experience building systems across databases, applied ML, and compute-heavy workloads. Led Huawei's learned query optimizer toward enterprise DBMS integration. Seeking roles where deep systems work becomes reliable software used at scale.

PROFESSIONAL EXPERIENCE

[Independent Engineering](#) & PhD Direction Evaluation (2025 – Present) | Remote

- Built a first-principles LLM [cost model](#) that predicted peak GPU memory and compute cost (FLOPs) within **~1–2%**, helping choose feasible model, batch, and hardware configurations without slow trial-and-error GPU runs.
- [Optimized](#) transformer training, reducing peak memory consumption by **3.8×**, and forward/backward time by **4.0×** in benchmarked runs, enabling up to **16×** higher sample-processing throughput *on the same hardware*.
- Built LLM benchmarks to isolate and evaluate techniques for *improving infrastructure efficiency*, demonstrating up to **5×** [higher](#) inference throughput, **2×** [lower](#) GPU memory use, and a **~2×** [increase](#) in maximum supported input length.

Huawei (2022 – 2025) | R&D Engineer, ML & DB | Russia, Novosibirsk (on-site)

- Led [HERO](#) project, a *learned* query optimizer for Huawei's enterprise DBMS, automating manual DBA tuning while delivering **3.2×** greater latency improvement than the literature SOTA and reducing regressed queries from **10%+ to 0%**.
- Resolved a key training-time bottleneck by independently selecting and adapting a research approach into a product-compatible pipeline, validating it, and driving its integration, ultimately reducing training time by **10×**.
- Designed and developed a replay-based benchmark framework that cut product-level experiment cycles from **~40 minutes** to **~1 second**, enabling rapid algorithm selection and de-risking optimizer development under uncertain requirements.
- Identified and resolved a critical optimizer inference bottleneck by designing a new inference pipeline, achieving **6×** faster inference and doubling end-to-end query-latency improvement from **~30%** to **~60%**.
- Proposed removing the neural network from the optimizer architecture and replacing it with an interpretable, reliable model, unblocking productization and *avoiding* a heavy PyTorch *dependency* inside the database kernel.

HSE University, HDI Lab (2021-2022) | Deep Learning Researcher | Russia, Moscow (remote)

- Developed [Star Algorithm](#), a data-efficient neural-network ensembling method that matched competitive ensemble baselines while providing *theoretical guarantees* on accuracy and near-optimal asymptotic convergence.
- Owned the full research-to-implementation cycle: formulated the algorithm, implemented PyTorch experiments, compared against common ensembling baselines, and *published* the result in Neural Networks and at an ICML Workshop.

SELECTED PUBLICATIONS

- “HERO: Hint-based efficient and reliable query optimizer”, S.Zinchenko, S.Iazov ([arXiv, 2024](#)).
- “Star algorithm for neural network ensembling”, S.Zinchenko, D.Lishudi ([Neural Networks, Q1, 2024](#); [ICML Workshop, 2022](#)).

EDUCATION

Novosibirsk State University — [QS Top 120 in Mathematics during studies](#)

- **MSc** in Applied Mathematics, with honors | Thesis: Hint-based query optimization for databases | GPA: **4.96 / 5.0**
- **BSc** in Mathematics and Computer Science, with honors | Thesis: Neural network ensemble algorithm | GPA: **4.98 / 5.0**

ACHIEVEMENTS & ACTIVITIES

- [Gold Medalist](#), International Olympiad in Cryptography, NSUCRYPTO, 2023
- “Employee of the Year”, Huawei, 2023
- 1st Degree, top 27 / 1,700+, National Student Math Olympiad “I Am Professional”, Yandex & MIPT, 2022
- Speaker at [PostgreSQL Bootcamp 2023 \(master class\)](#), [ISP RAS 2024 \(technical talk\)](#), and [SSOPT 2024 Summer School](#) (lecture)
- Reviewed research papers for DASFAA 2023 & 2024 (Springer Nature)

TECHNICAL SKILLS

- **Basics:** Python, SQL, Git, Linux; **ML:** PyTorch, scikit-learn, NumPy, Pandas, clearML; **Backend / Tooling:** PostgreSQL, Pydantic, pytest, GitHub Actions CI; **LLM Training/Fine-tuning:** HuggingFace, SFT, RLHF, GRPO, PEFT, LoRA, Quantization; **High-performance / Distributed Compute:** torch.distributed, torch.compile, FSDP, Triton;